The Higher-Dimensional Spin-Ball/Max-Ball Question Has a Negative Answer

fa_banach_001 / agent_lane_00

17 August 2026

Status: literature already answered (full answer for every $m \geq 3$).

Original question

Farenick, Huntinghawk, Masanika, and Plosker prove that their spin ball and max ball agree in dimensions one and two. Immediately after Theorem 5.12, on PDF page 19, they ask whether the theorem extends to higher dimensions [1].

The corrected universal three-variable coefficient tuple is

$$F^{[3]} \cong P \oplus \overline{P}, \qquad \mathcal{D}_{F^{[3]}} = \mathcal{D}_P \cap \mathcal{D}_{\overline{P}}.$$

This point matters: testing only the irreducible Pauli triple P produces a spurious level-two "counterexample" that merely detects the failure of the transpose map to be completely positive. It is excluded by the conjugate summand in the corrected universal tuple.

Explicit later answer

Evert and Passer restate the same problem as Question 1 on PDF page 6, explicitly citing the text after Farenick et al.'s Theorem 5.12 [2]. Their Theorem 2.9 (PDF page 10) constructs free extreme points of $\mathcal{D}_{F^{[3]}}$ at matrix levels four and six. Hence $\mathcal{D}_{F^{[3]}}$ is not the minimal matrix convex set over the Euclidean ball, and its free polar dual is not maximal. Remark 2.11 (PDF page 11) states directly that this gives a negative answer to the spin-ball/max-ball equality question in three or more variables (accounting for the reversed min/max terminology in the source paper).

Finally, Corollary 3.9 (PDF page 19) proves for every $g \geq 3$ that $\mathcal{D}_{F^{[g]}}$ has a higher-level free extreme point and therefore is not the minimal matrix convex set over $\mathbb{B}_g$. Thus the proposed equality fails in every higher dimension, not merely for $g = 3$.

The supporting authors knew they were resolving the source question. This is therefore an explicit literature answer rather than an agent-inferred reformulation.

Scope and search evidence

The source question is fully resolved: equality holds for $m = 1, 2$ and fails for every $m \geq 3$. The later paper was found by checking the citation graph of arXiv:2006.06810 and then matching its Question 1 and Remark 2.11 against the exact source location. No new proof or counterexample is claimed here.

References

[1] D. Farenick, F. Huntinghawk, A. Masanika, and S. Plosker, *Complete order equivalence of spin unitaries*, Linear Algebra Appl. **610** (2021), 1–28; arXiv:2006.06810. Corrigendum: Linear Algebra Appl. **635** (2022), 201–202.

[2] E. Evert and B. Passer, *Matrix convex sets over the Euclidean ball and polar duals of real free spectrahedra*, arXiv:2507.20325 (2025), doi:10.1016/j.laa.2026.01.015.